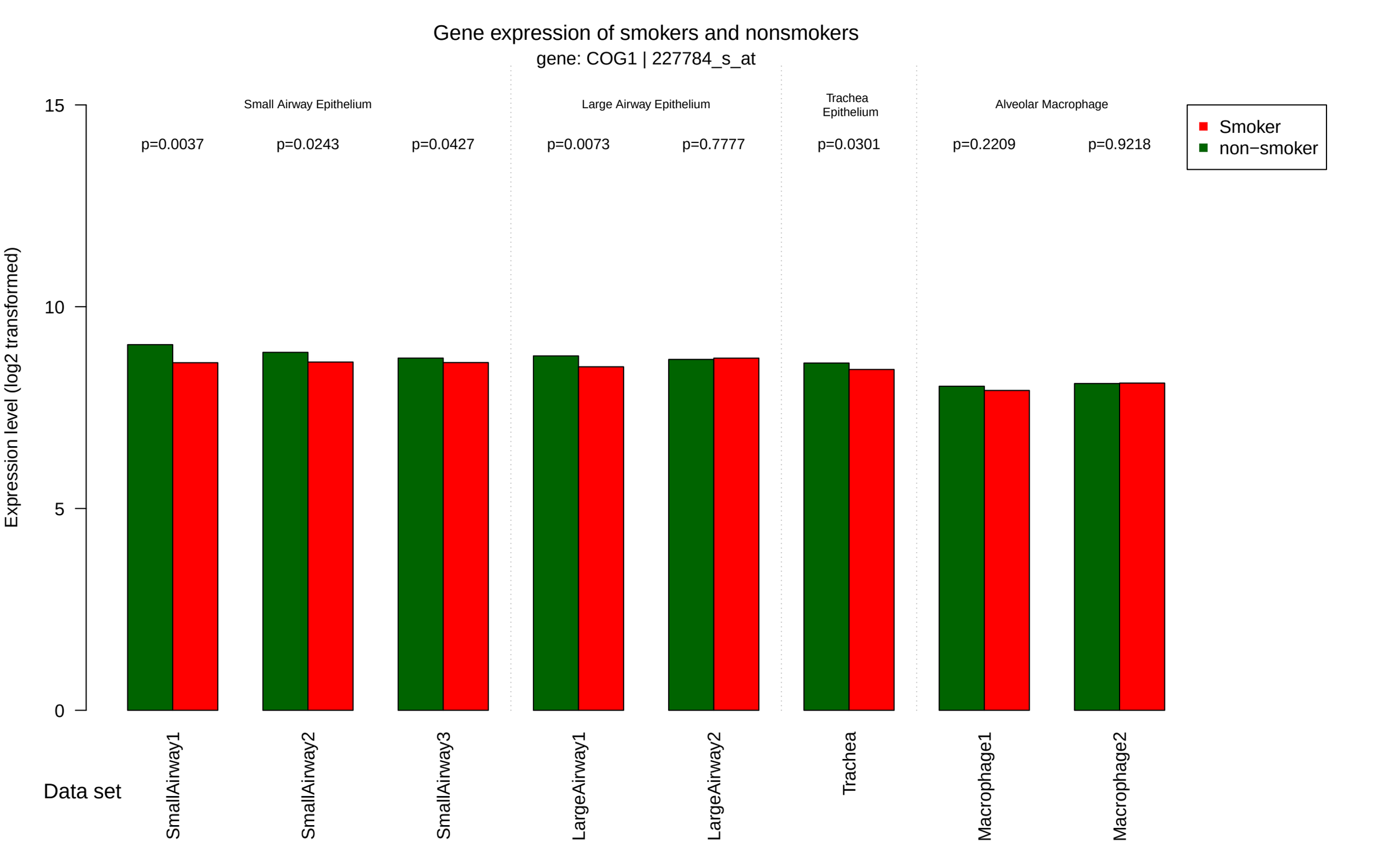
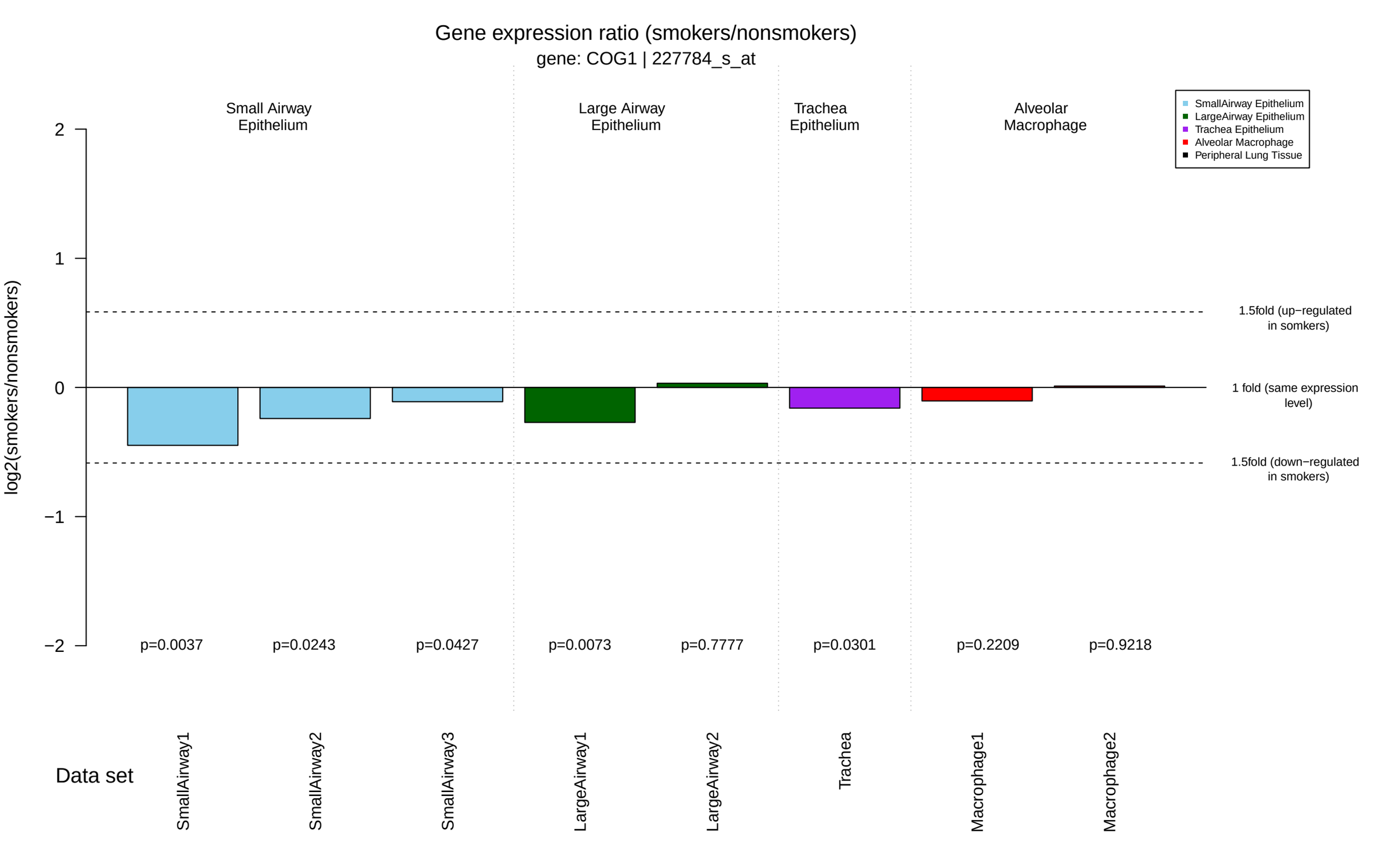
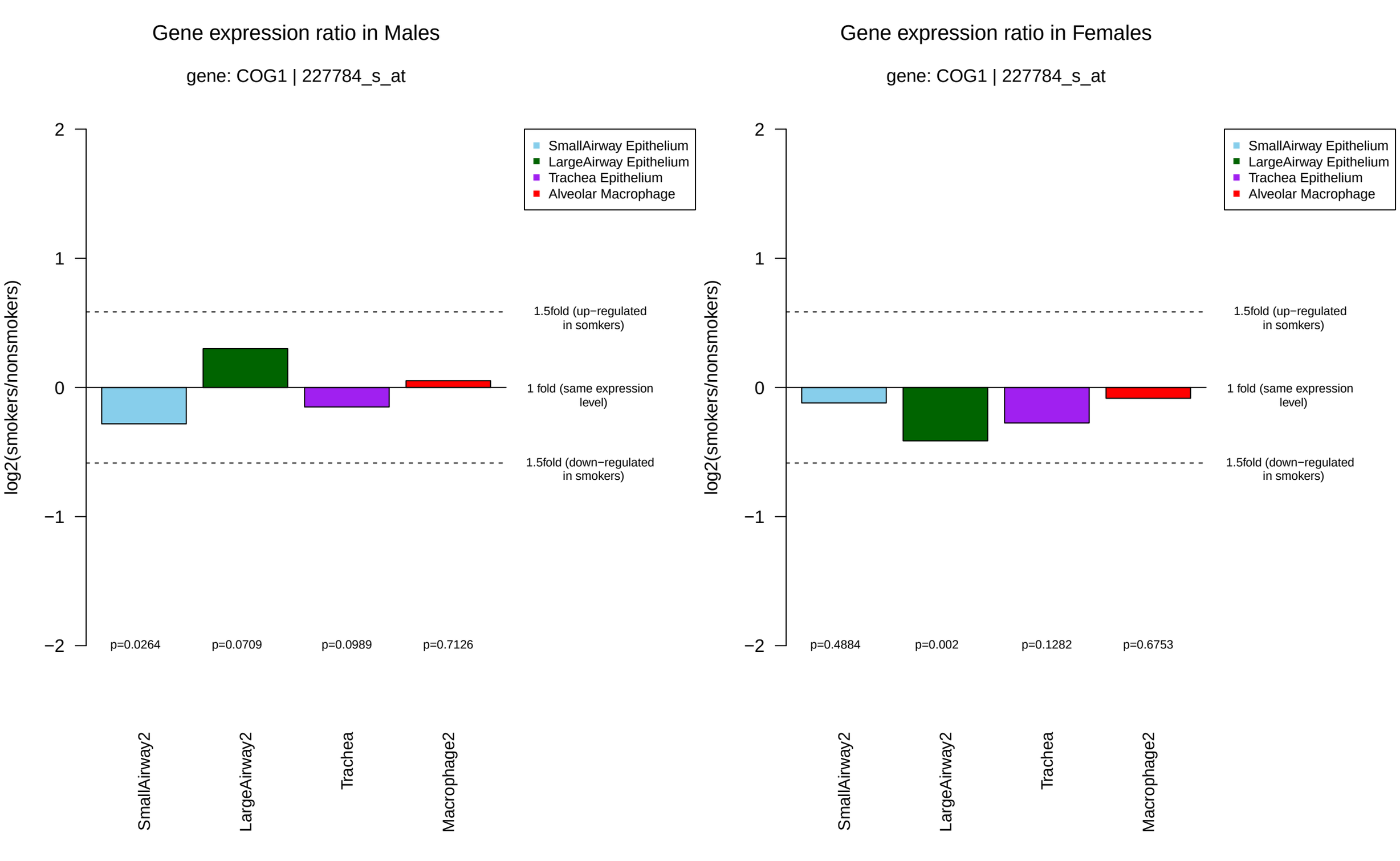
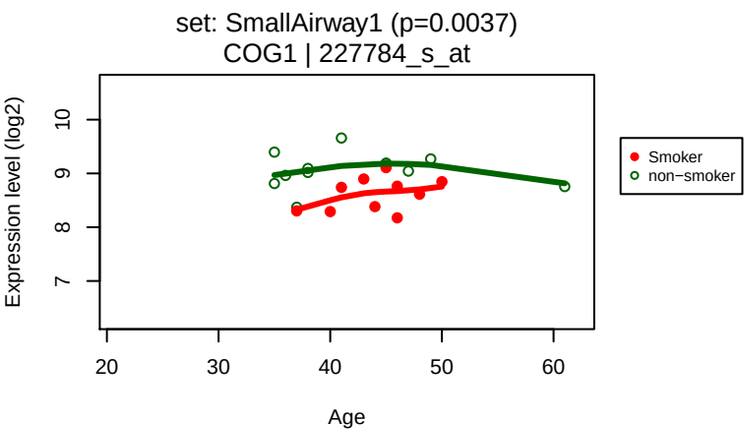
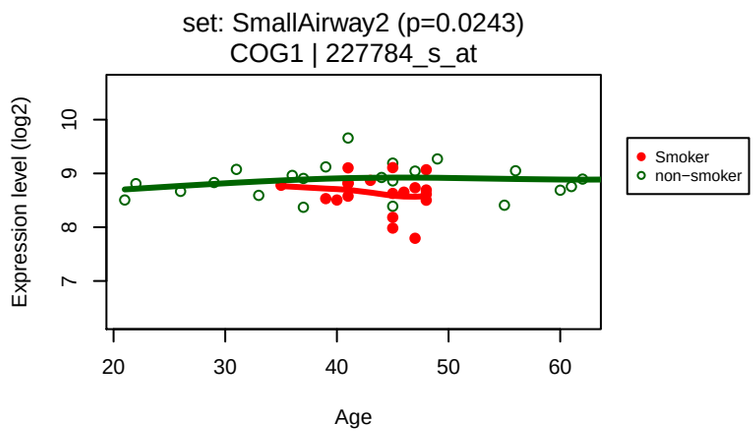
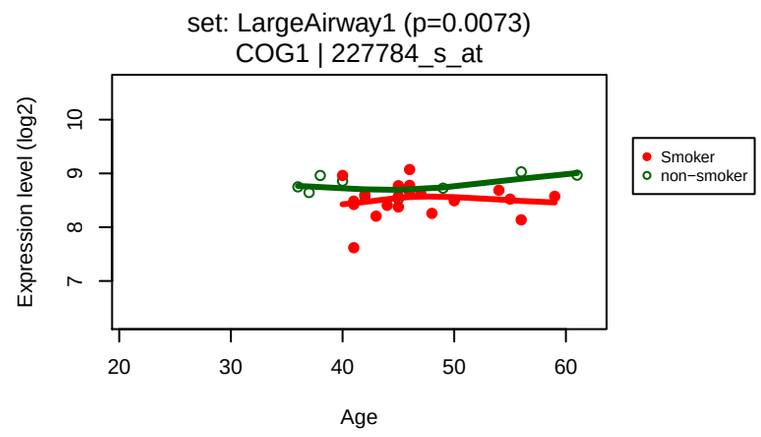

Gene expression of smokers and nonsmokers
gene: COG1 | 227784_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0037
p=0.0243
p=0.0427
p=0.0073
p=0.7777
p=0.0301
p=0.2209
p=0.9218
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

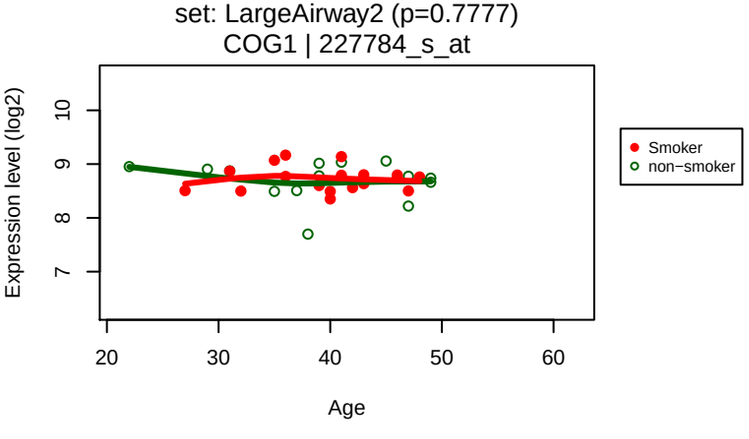

Gene expression ratio (smokers/nonsmokers)
gene: COG1 | 227784_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0037
p=0.0243
p=0.0427
p=0.0073
p=0.7777
p=0.0301
p=0.2209
p=0.9218
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

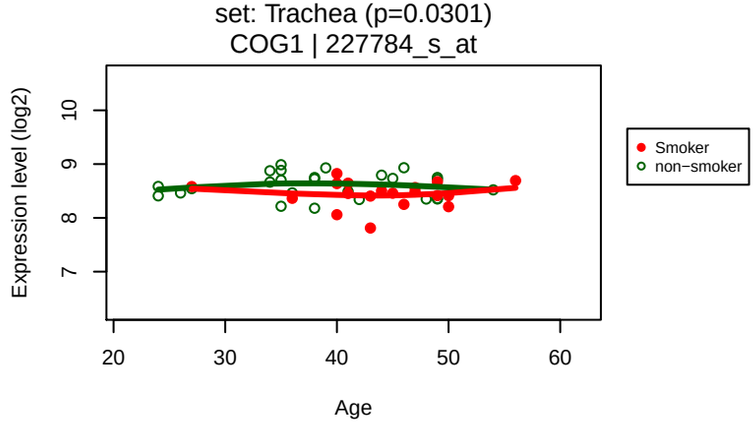

Gene expression ratio in Males
gene: COG1 | 227784_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0264
p=0.0709
p=0.0989
p=0.7126
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COG1 | 227784_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4884
p=0.002
p=0.1282
p=0.6753
SmallAirway2
LargeAirway2
Trachea
Macrophage2

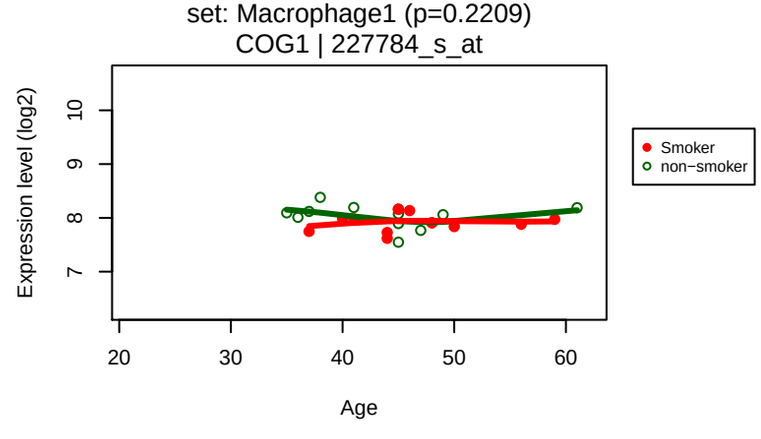

set: SmallAirway1 (p=0.0037)
COG1 | 227784_s_at
Expression level (log2)
Age
Smoker
non−smoker

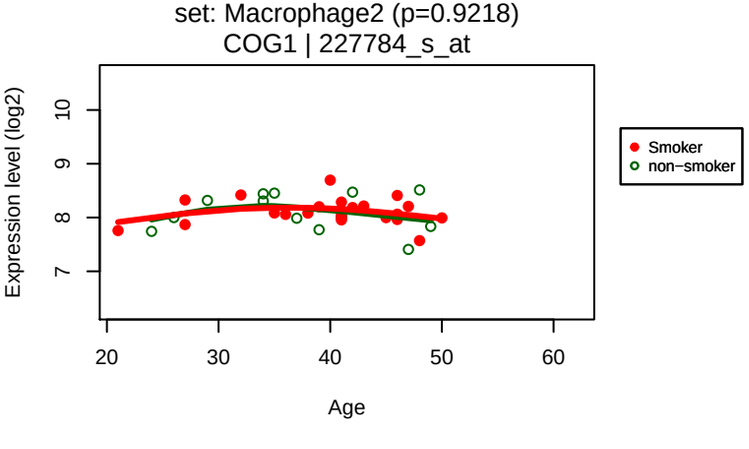

set: SmallAirway2 (p=0.0243)
COG1 | 227784_s_at
Expression level (log2)
Age
Smoker
non−smoker

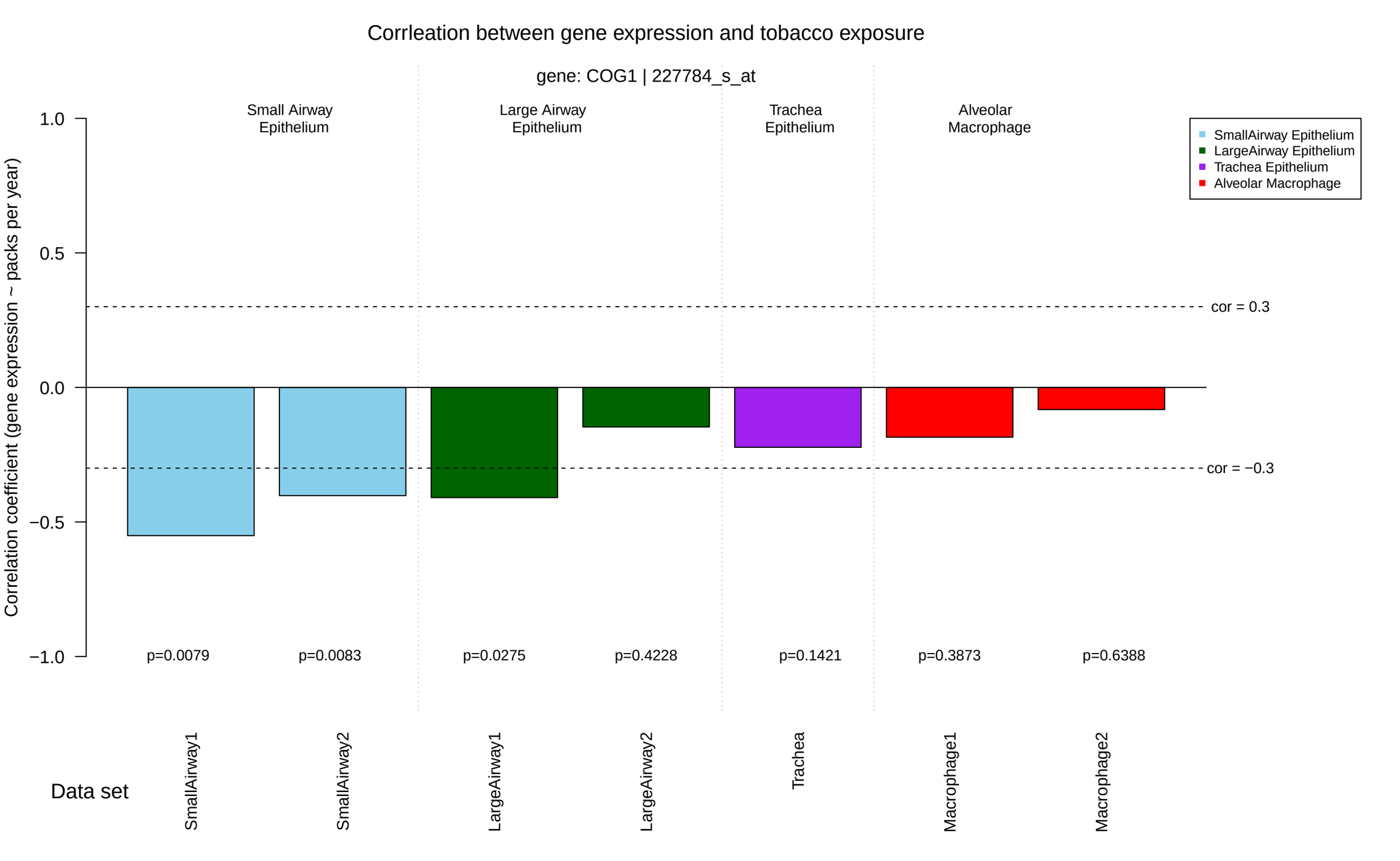

set: LargeAirway1 (p=0.0073)
COG1 | 227784_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7777)
COG1 | 227784_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0301)
COG1 | 227784_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2209)
COG1 | 227784_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9218)
COG1 | 227784_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COG1 | 227784_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0079
p=0.0083
p=0.0275
p=0.4228
p=0.1421
p=0.3873
p=0.6388
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2